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Tutorial 3

DSAI6707: Artificial Intelligence Concepts

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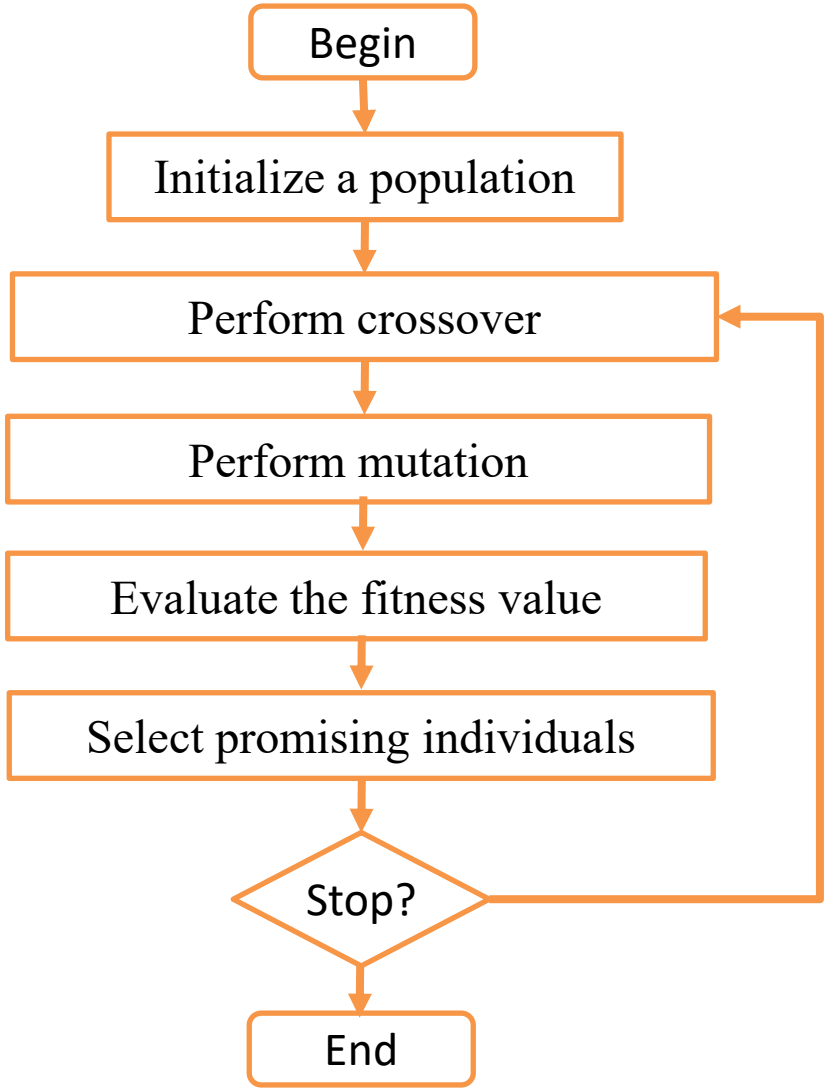
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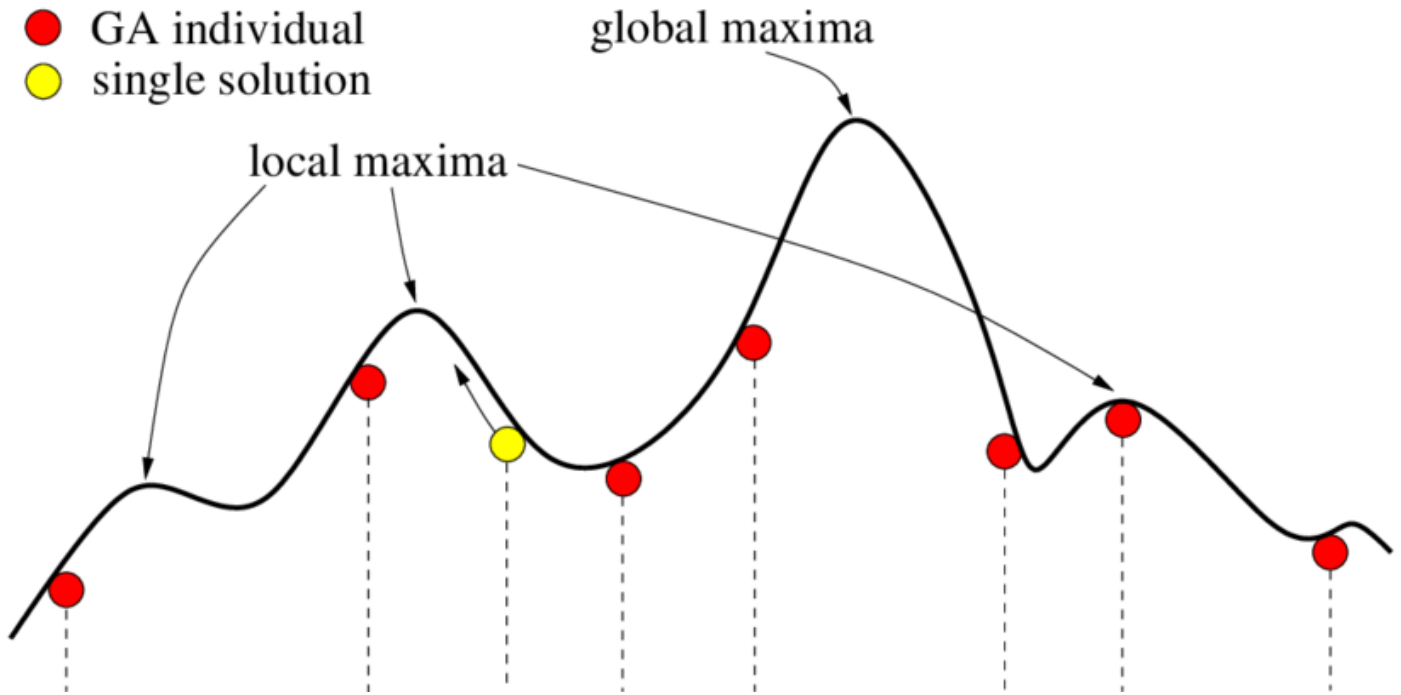
Genetic Algorithm

Framework

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Darwinian evolution theory: Survival of the fittest





Fitness landscape

Encoding

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Binary Encoding

Solution x: **0** Chromosome: **00000**

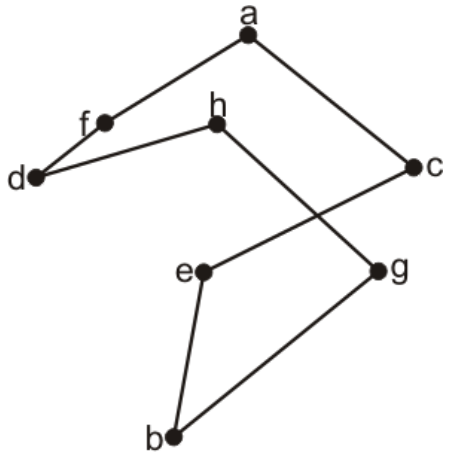
Solution x: **5** → Chromosome: **00101**

Solution x: **31** Chromosome: **11111**

Real-number Encoding

Solution x: **5** → Chromosome: **5**

Encoding dependent on specific problems, e.g. traveling salesman problem



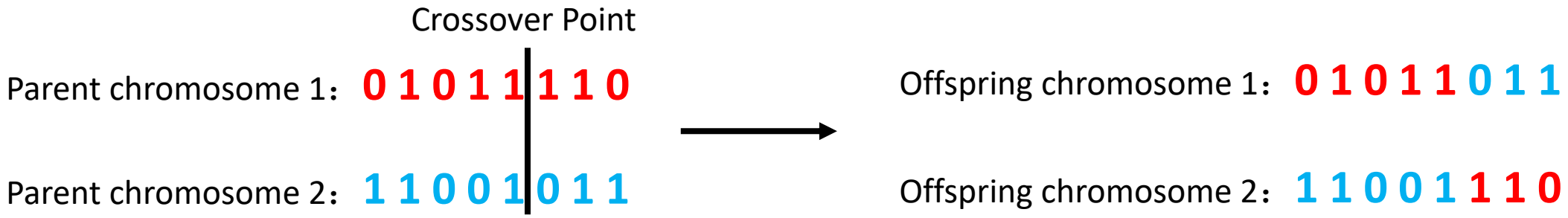
→

a	c	e	b	g	h	d	f

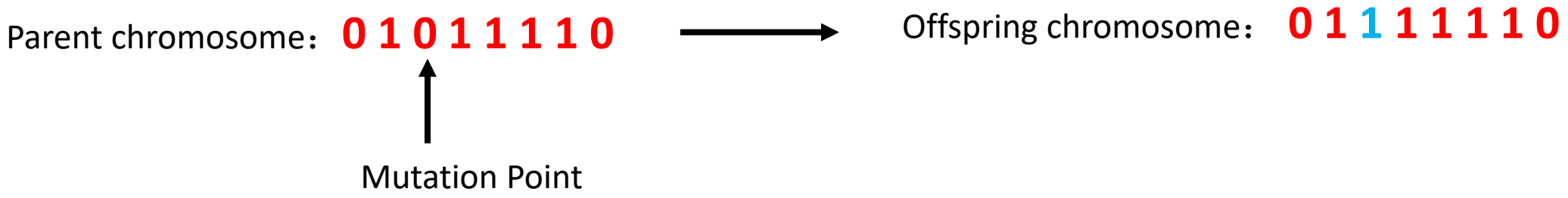
Crossover and Mutation

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Crossover:



Mutation:



Evaluate Fitness Value

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Chromosome 1： **0 0 1 1**

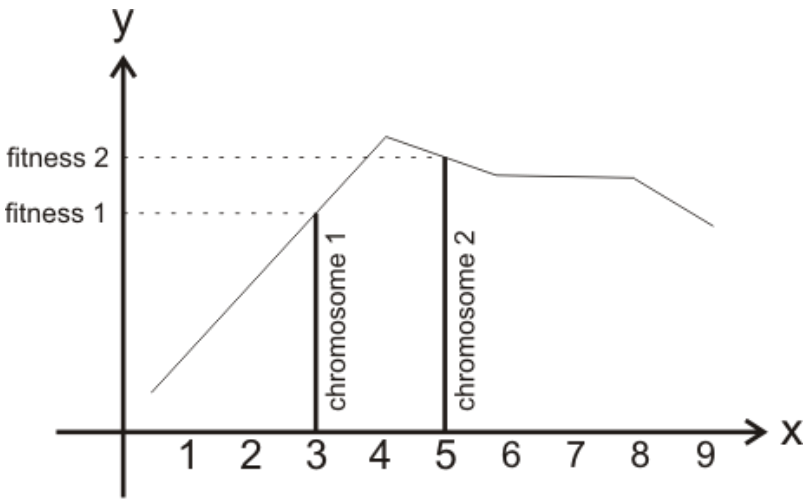
Chromosome 2： **0 1 0 1**

Decode →

Solution 1： **3**

Solution 2： **5**

Calculate fitness function →



Selection

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For maximizing a problem: $f(x)$, we have 3 individuals x_1, x_2, x_3 and their fitness values $f(x_1) = 15, f(x_2) = 5, f(x_3) = 80$

Rank Weighting Selection:

The probability that an individual in the population is selected is proportional to the rank of the individual.

- The ranks of the 3 individuals are $rank(x_1) = 2, rank(x_2) = 1, rank(x_3) = 3$
- The probability of x_1 being chosen is $p(x_1) = ?, p(x_2) = ?, p(x_3) = ?$

Fitness Weighting Selection (Roulette Wheel Selection):

The probability that an individual in the population is selected is proportional to the fitness value of the individual.

- The probability of x_1 being chosen is $p(x_1) = ?, p(x_2) = ?, p(x_3) = ?$

Selection

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For maximizing a problem: $f(x)$, we have 3 individuals x_1, x_2, x_3 and their fitness values $f(x_1) = 15, f(x_2) = 5, f(x_3) = 80$

Rank Weighting Selection:

The probability that an individual in the population is selected is proportional to the rank of the individual.

- The ranks of the 3 individuals are $rank(x_1) = 2, rank(x_2) = 1, rank(x_3) = 3$
- The probability of x_1 being chosen is $p(x_1) = \frac{rank(x_1)}{rank(x_1)+rank(x_2)+rank(x_3)} = \frac{1}{3}, p(x_2) = \frac{1}{6}, p(x_3) = \frac{1}{2}$

Fitness Weighting Selection (Roulette Wheel Selection):

The probability that an individual in the population is selected is proportional to the fitness value of the individual.

- The probability of x_1 being chosen is $p(x_1) = \frac{f(x_1)}{f(x_1)+f(x_2)+f(x_3)} = 0.15, p(x_2) = 0.05, p(x_3) = 0.8$

Selection

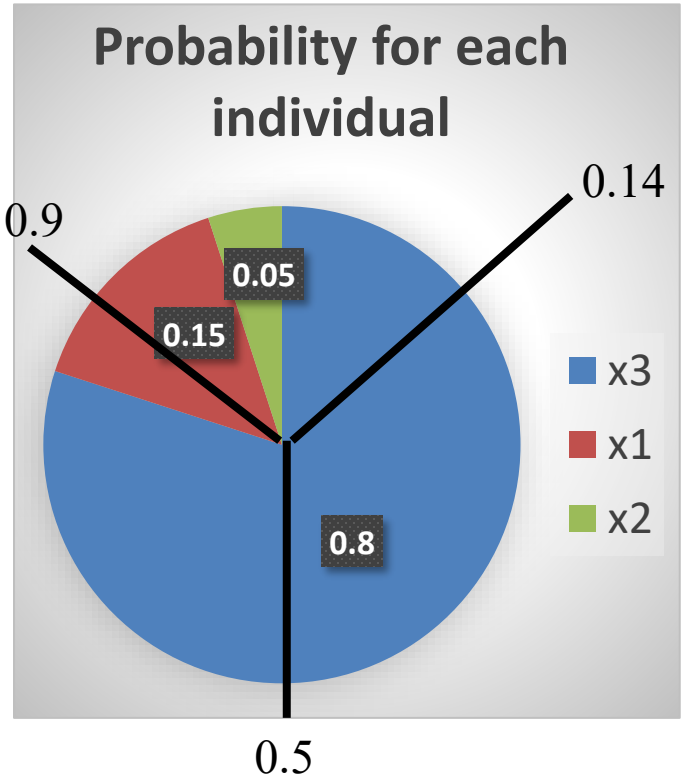
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For maximizing a problem: $f(x)$, we have 3 individuals x_1, x_2, x_3 and their fitness values $f(x_1) = 15, f(x_2) = 5, f(x_3) = 80$

If 3 individuals should be selected from x_1, x_2, x_3, which 3 individuals will be selected? Take the fitness weighting selection as an example:

- The probability of x_1 being chosen is $p(x_1) = \frac{f(x_1)}{f(x_1)+f(x_2)+f(x_3)} = 0.15, p(x_2) = 0.05, p(x_3) = 0.8$
- Obtain the probability table.

Chromosome	Fitness	Probability
x_3	80	0.8
x_1	15	0.15
x_2	5	0.05



- Since 3 individuals should be selected, we can generate 3 random numbers which are uniformly sampled from [0,1], assume these 3 random numbers are 0.14, 0.5, and 0.9.
- We can draw a pie chart as the right figure to simulate the selection process.
- Random number 0.14 points the bule region of x_3, so x_3 is selected.
- Random number 0.5 points the bule region of x_3, so x_3 is selected again.
- Random number 0.9 points the bule region of x_3, so x_1 is selected.

Exercise

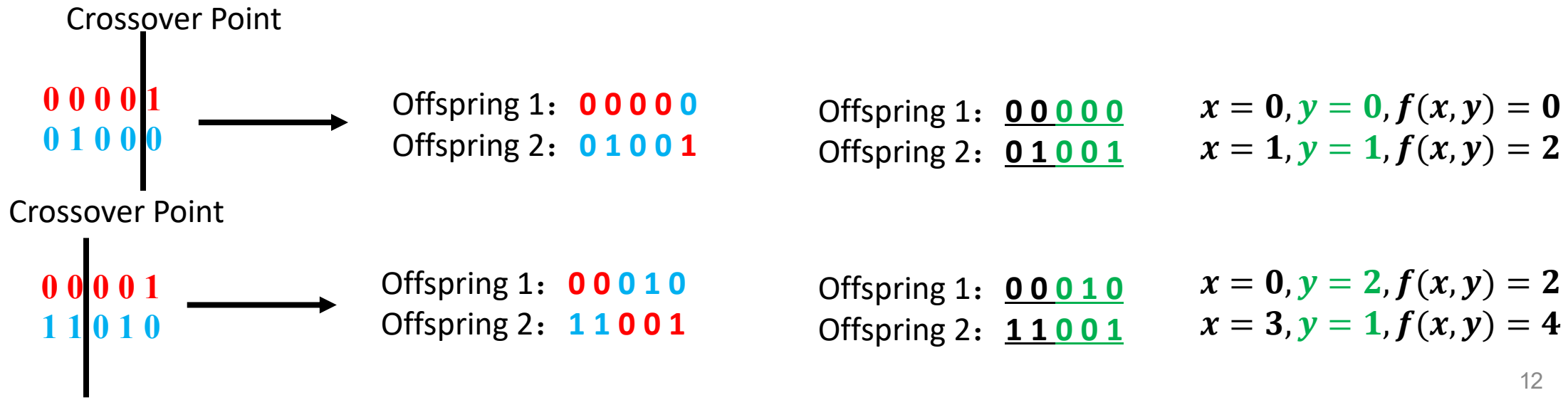
Exercise 1

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Three chromosomes 00001, 01000, 11010 are initialized for maximizing $f(x, y) = x + y$. x is represented by the first 2 bits and y is represented by the last 3 bits.
1-st and 2-nd chromosomes are selected for crossover and the crossover point is between 4-th point and 5-th point.
1-st and 3-rd chromosomes are selected for crossover and the crossover point is between 2-nd point and 3-rd point.
Please calculate the fitness values of the offspring.

Exercise 1

Three chromosomes 00001, 01000, 11010 are initialized for maximizing $f(x, y) = x + y$. x is represented by the first 2 bits and y is represented by the last 3 bits.
1-st and 2-nd chromosomes are selected for crossover and the crossover point is between 4-th point and 5-th point.
1-st and 3-rd chromosomes are selected for crossover and the crossover point is between 2-nd point and 3-rd point.
Please calculate the fitness values of the offspring.



Exercise 2

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For the binary genetic algorithm with 4 individuals. The 4 individuals and their fitness values for a **maximization** problem are presented in the Table.

Solution	Chromosome	Fitness
1	01101	169
2	11000	576
3	01000	64
4	10011	361

Q1:

(a) Obtain the probability table for **rank weighting selection**.

(b) If 2 individuals should be selected, and the corresponding 2 random numbers generated are 0.15, 0.82, which two chromosomes will be selected?

Exercise 2

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Answer for Q1: (a) Obtain the probability table for rank weighting selection.

Solution	Chromosome	Fitness	Rank	Probability
2	11000	576	4	4/(1+2+3+4)=0.4
4	10011	361	3	3/(1+2+3+4)=0.3
1	01101	169	2	2/(1+2+3+4)=0.2
3	01000	64	1	1/(1+2+3+4)=0.1

Exercise 2

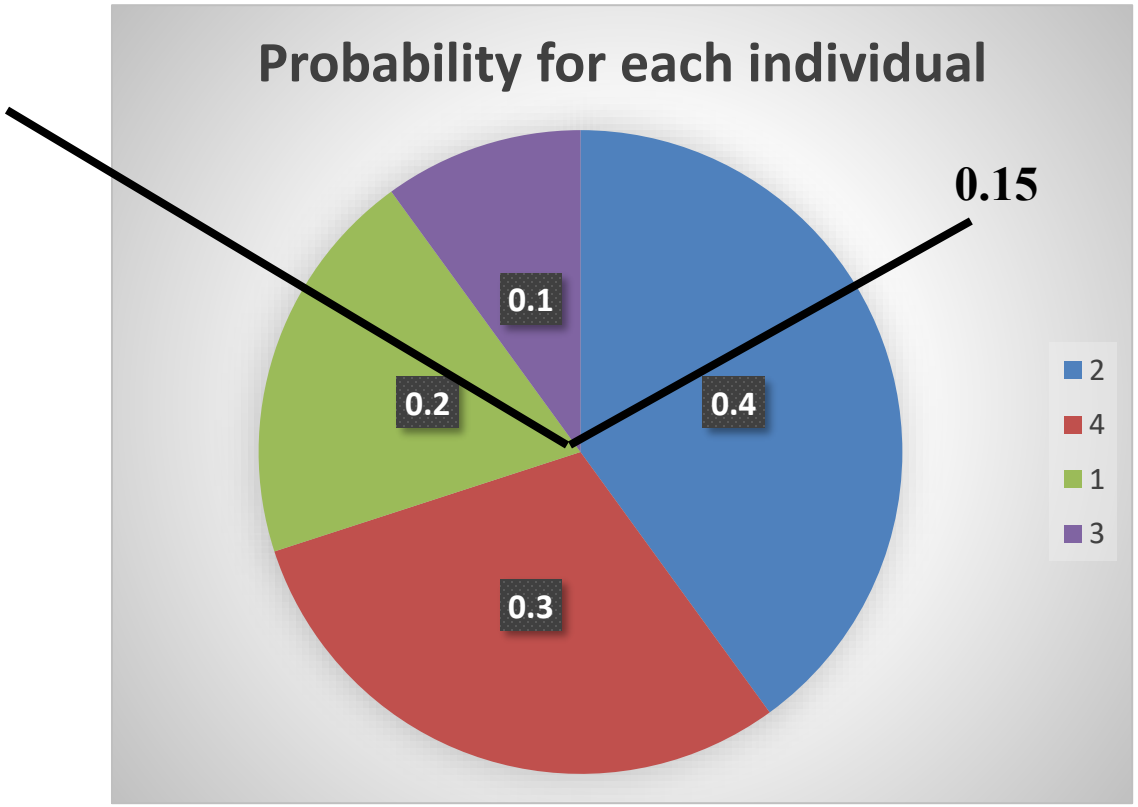
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Answer for Q1:

(b) If 2 individuals should be selected, and the corresponding 2 random numbers generated are 0.15, 0.82, which two chromosomes will be selected?

Solution	Chromosome	Fitness	Rank	Probability
2	**11000**	**576**	**4**	**0.4**
4	10011	361	3	0.3
1	**01101**	**169**	**2**	**0.2**
3	01000	64	1	0.1

Chromosomes 2 and 1 are selected.



Exercise 2

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For the binary genetic algorithm with 4 individuals. The 4 individuals and their fitness values for a **maximization** problem are presented in the Table.

Solution	Chromosome	Fitness
1	01101	169
2	11000	576
3	01000	64
4	10011	361

Q2:

(a) Obtain the probability table for **fitness weighting selection**.

(b) If 2 individuals should be selected, and the corresponding 2 random numbers generated are 0.15, 0.82, which two chromosomes will be selected?

Exercise 2

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Answer for Q2: **(a)** Obtain the probability table for fitness weighting selection.

Solution	Chromosome	Fitness	Probability
2	11000	576	576/(169+576+64+361)=0.49
4	10011	361	361/(169+576+64+361)=0.31
1	01101	169	169/(169+576+64+361)=0.14
3	01000	64	64/(169+576+64+361)=0.06

Exercise 2

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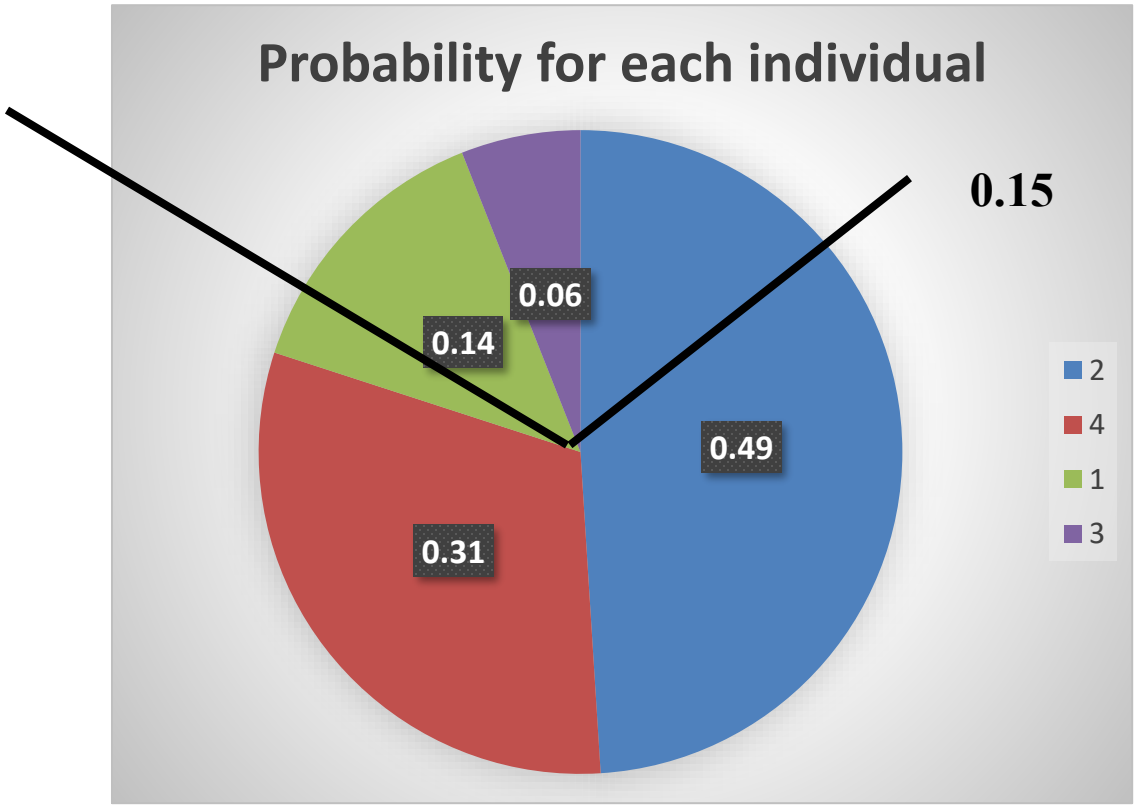
Answer for Q2:

(b) If 2 individuals should be selected, and the corresponding 2 random numbers generated are 0.15, 0.82, which two chromosomes will be selected?

0.82

Solution	Chromosome	Fitness	Probability
2	11000	576	0.49
4	10011	361	0.31
1	01101	169	0.14
3	01000	64	0.06

Chromosomes 2 and 1 are selected.



Exercise 3

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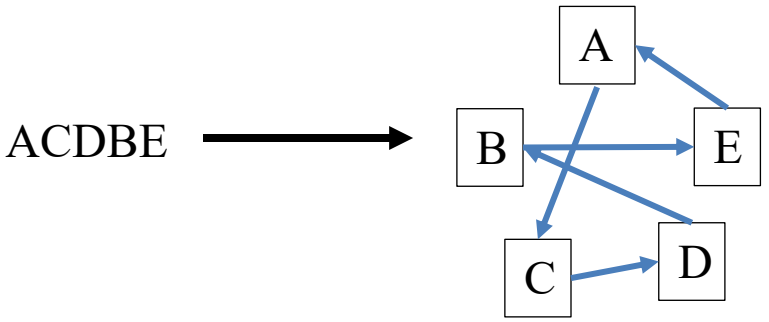
Two chromosomes ACDBE, ADECB are initialized for solving a traveling salesman problem. The distance matrix of the five nodes is given in the table.

	A	B	C	D	E
A	0	56	31	8	10
B	56	0	9	40	7
C	31	9	0	15	20
D	8	40	15	0	5
E	10	7	20	5	0

The mutation operator is to exchange the positions of 2 mutation points.

- For chromosome ACDBE, the mutation points are 2-rd point and 4-th point.
- For chromosome ADECB, the mutation points are 5-th point and 4-th point.

If two chromosomes should be selected **from parents and offspring** by **rank weighting selection**, which two chromosomes will be selected (The random numbers are 0.3, 0.6)?



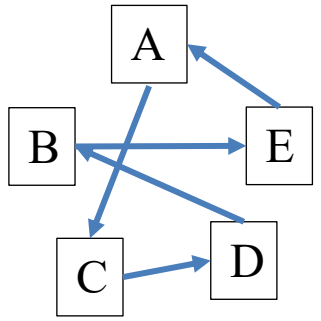
Exercise 3

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Two chromosomes ACDBE, ADECB are initialized for solving a traveling salesman problem. The distance matrix of the five nodes is given in the table. (The random numbers are 0.3, 0.6)

	A	B	C	D	E
A	0	56	31	8	10
B	56	0	9	40	7
C	31	9	0	15	20
D	8	40	15	0	5
E	10	7	20	5	0

ACDBE →



Parents:

- ACDBE, Cost=AC+CD+DB+BE+EA=31+15+40+7+10=103,Rank=2
- ADECB, Cost=AD+DE+EC+CB+BA=8+5+20+9+56=98,Rank=3

Offspring:

- ABDCE, Cost=AB+BD+DC+CE+EA=56+40+15+20+10=141,Rank=1
- ADEBC, Cost=AD+DE+EB+BC+CA=8+5+7+9+31=60,Rank=4

Chromosome	Cost	Probability
ADEBC	60	4/(4+3+2+1)=0.4
ADECB	98	3/(4+3+2+1)=0.3
ACDBE	103	2/(4+3+2+1)=0.2
ABDCE	141	1/(4+3+2+1)=0.1

Chromosomes ADEBC and ADECB are selected.

Exercise 4

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By considering the individual and combined effect of reproduction, crossover and mutation in genetic algorithms, derive the schemata theorem to examine the growth rates of various schemata or similarity templates contained in a population. The schemata theorem is given as,

$$n_s(t+1) \approx n_s(t) \cdot \frac{fit(S)}{< fit >} \left(1 - P_{cross} \frac{d(S)}{l-1} - P_m o(S) \right)$$

A population contains the following strings and fitness values at generation 0:

#	String	Fitness
1	10001	20
2	11100	10
3	00011	5
4	01110	15

The probability of mutation is $P_m = 0.01$ and the probability of crossover is $P_{cross} = 1$. Using the schemata theorem to estimate the expected number of schemata of the form 1**** in generation 1.

Exercise 4

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From the schemata theorem,

$$n_s(t+1) \approx n_s(t) \cdot \frac{fit(S)}{< fit >}\left(1 - P_{cross}\frac{d(S)}{l-1} - P_m o(S)\right)$$

The number of schemata at generation 0 is 2, i.e., $n_s(t) = 2$ and the length of the string $l = 5$.

For schema of 1****, the defining length $d(S) = 0$, the order $o(S) = 1$, $fit(S) = 15$ and $<fit> = 50/4$. Hence the expected number of schemata at generation 1 is given as,

$$n_s(1) = 2 \cdot \frac{15(4)}{50}(1 - 0 - 0.01) = 2.376$$

Thanks for your time